

Volume word problem special project

water displacement method · overflow problem · irregular volume · comprehensive application · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5B Unit 5 + Modern Education 5L B Unit 16

Core Traps: □ T1 Volume Operations - Forgetting Unit Conversion · T4 Geometry Application - Confusing water level rise with object volume

SSPA Association: ● Highest frequency Must be tested in paper 1/paper 2, accounting for about 70% of volume application questions. It is the most important geometry application course in P5

Prerequisite knowledge: L26 (cuboid/cube volume formula · base area calculation · capacity unit conversion $1\text{L} = 1000\text{ cm}^3$)

Class objectives: ❶ Master the principles of drainage method ❷ Deal with overflow problems ❸ Measure the volume of irregular objects ❹ Multi-step container comprehensive questions

Student Name:

Class:

Date:

Time Spent:

I.Warm-Up Questions (total 5 question · 5 minutes)

#	Question	Difficulty	Working Space (Show full working)
1	A cuboid is 8 cm long, 5 cm wide, and 3 cm high. Volume = ?	Basic	
2	A cube, side length 4 cm, volume = ?	Basic	
3	1 liter = _____ cubic centimeter (cm ³)	Basic	
4	A rectangular container has base area = 40 cm ² and height 20 cm. What is the total capacity of the container in cm ³ ? How many L is it?	Basic	
5	There is water in a cup. A pebble is placed in the cup and the water level rises. Why does the water surface rise? (Explained using "volume" and "space")	Advanced	

II.Core Knowledge + Worked Examples

Knowledge point 1: Basics of drainage method - the secret of rising water surface SSPA must take the exam

- ① Principle of drainage method |||SEP|||: When an object is completely immersed in water, it will displace the same volume of water → Water surface rises
Core formula °
- ② Object volume = Container bottom area × Height of water surface rise : Water surface rise height = Object volume ÷ Container bottom area
- ③ New water height = Original water height + Water surface rising height
- ④ Prerequisites |||SEP|||: The object must be
- ⑤ completely submerged |||SEP|||. If the object is not completely submerged, the drainage method formula does not apply! The object must fully immersed. If the object is not completely submerged, the drainage method formula does not apply!



Trap detonation example - volume vs water height

Example 1: The bottom area of a rectangular container is 50 cm², and the original water height is 8 cm. After placing a stone (completely submerged), the water level rises to 11 cm. What is the volume of the stone?

 **Common mistakes (60% students)**

 **Correct solution**

$$\text{Volume} = 11 \times 50 = 550 \text{ cm}^3$$

Use the new water level $\times$ bottom area directly, without counting the "rising part"

$$\text{rise} = 11 - 8 = 3 \text{ cm}$$

$$\text{Volume} = 50 \times 3 = 150 \text{ cm}^3$$

volume = basearea $\times$ water level "change" (increase in height)

example

Example 2: A container has a bottom area of 60 cm^2 and an iron block with a volume of 180 cm^3 is placed (completely submerged). How many centimeters did the water surface rise? What is the new water height? (raw water height 5 cm)

①

Find the bottom area

Length $\times$ Width or known value

②

Calculate water level changes

Rising = volume $\div$ bottom area
falling = removal $\div$ bottom area

③

Calculate new water level

New = original + rising
new = original - falling

④

verification unit

$\text{cm}^3 \leftrightarrow \text{mL} \leftrightarrow \text{L}$ is calculated only if it is consistent



Tip: "When an object is immersed in water, the water level will rise. The volume is equal to the base area, multiplied by the increase - remember it's a 'change' not a 'total height'!"

⚠ The first high-frequency error: using the new water height $\times$ the base area as the volume of the object. Volume = bottom area $\times$ (new water level - original water level)!

⚠ The second most common mistake: forgetting to verify that the object is fully submerged. If the object floats or comes out of the water, the drainage method is not valid.

Knowledge Point 1 Synchronization practice (write out ①basearea ②water level change ③object volume)

#	Question	Difficulty	Working Space
6	The bottom area is 30 cm^2 , and the water surface rises by 5 cm after placing the object. Object volume = ?		
7	The base area is 45 cm^2 and the raw water height is 10 cm. Completely submerged after placing an object with a volume of 225 cm^3 . How much does the water rise? How high is the new water?		
8	The rectangular container is 20 cm long, 15 cm wide, and the raw water is 12 cm high. After placing the stones the water height rose to 15 cm. Volume of stone = ?		

9	The side length of the cube container is 10 cm and the height of the water is 6 cm. Place a metal block with a volume of 300 cm ³ (completely submerged). What is the new water height?		
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Knowledge point 2: Overflow problem - when the object is too big, the water will overflow SSPA Advanced

- ① **Overflow condition** |||SEP|||: Input volume > Container Remaining space (empty space). Remaining space
 ② = total container capacity – original volume of water. Overflow volume
 ③ = input volume – remaining space. ④ If the volume of the input object is ≤ the remaining space, then will not overflow and will only cause the water level to rise (return to KP1). Unit conversion
 ⑤ **Unit conversion** : 1 L = 1000 cm³ · 1 mL = 1 cm³ ·



Trap examples

Example 3: The total capacity of a container is 2000 cm³, and the original water is 1800 cm³. Place a stone with a volume of 500 cm³ (completely submerged). Will the water overflow? If so, how much cm³ is spilled?

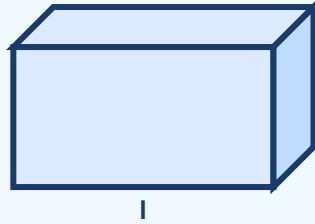
example

Example 4: The capacity of a fish tank is 50 L, and the original water is 45 L. Place a rockery with a volume of 8 L. Will the water overflow? Why?

Knowledge Point 2 Synchronization practice

#	Question	Difficulty	Working Space
10	Container capacity 3 L, original water 2.5 L (i.e. 2500 cm ³). An object with a volume of 800 cm ³ is placed. (a) Remaining space = ? (b) Will it overflow? How much overflow?		
11	Container capacity 1000 cm ³ , original water 750 cm ³ . An object with a volume of 200 cm ³ is placed. Will it overflow?		
12	The fish tank capacity is 50 L, and the original water is 45 L. Place a rockery with a volume of 3 L. Will it overflow? How many L is spilled?		
13	A cube container has an unknown side length and a capacity of 2.4 L. Original water 2 L. What is the maximum volume of objects that can be put in without overflowing? (Answer in cm ³)		

 Standard cuboid container reference picture



Length (l) $\times$ width (w) $\times$ height (h). Base area = $l \times w$. Capacity = $l \times w \times h$. Remaining space = capacity - original water volume. Overflow = input volume - remaining space.

Knowledge Point 3: Volume of Irregular Objects - Shapes without formulas also have energy **SSPA must take the exam**

① **Irregular objects** (such as stones, keys, potatoes) cannot calculate the volume using length $\times$ width $\times$ height $\rightarrow$ use **drainage method**.

② **Measurement steps** |||SEP|||: ① **Record the original water level** ② **Place the object (completely submerged)** ③ **Record the new water level** ④ **Volume = bottom area $\times$ (new water level - original water level)**. Measuring cup/cylinder |||SEP|||: Usually cylindrical, the base area is known or can be calculated. The difference in water levels is the change in height.

③ ④ **If the object floats** on the water surface (density is less than water), the drainage method is not applicable - other methods need to be used (such as pressing into the water). **float on water** (Denser than water), drainage method is not applicable - other methods need to be used (such as pressing water in).



example

Example 5: The bottom area of the measuring cup is 20 cm^2 , and the original water height is 15 cm. Add a pellet (completely submerged) and raise the water height to 16 cm. What is the volume of the wave?

example

Example 6: The bottom area of a rectangular water tank is 200 cm^2 , and the water height is 25 cm. After placing an irregular stone, the water height rose to 28 cm. What is the volume of the stone in cm^3 ?

Knowledge Point 3 Synchronization practice

#	Question	Difficulty	Working Space
14	The bottom area of the measuring cup is 15 cm^2 , the height of the water is 10 cm $\rightarrow$ after placing the object, it rises to 12 cm. Object volume = ?		

15	The bottom area of the rectangular water tank is 50 cm^2 , and the water level rises from 8 cm to 11 cm. What is the volume of the object thrown in?		
16	The bottom area of the measuring cup is 80 cm^2 , and the water level rises from 6 cm to 6.5 cm. Volume of the input object = ?		
17	An irregular metal block is placed in a measuring cup (base area 25 cm^2) and the water level rises from 20 cm to 23 cm. (a) Volume of metal block = ? (b) If the total height of the measuring cup is 25 cm and the original water is 20 cm high, what is the maximum volume it can hold without overflowing?		

Knowledge point 4: Comprehensive application - multi-step container problem

● SSPA killer question

① **Multi-object problem** |||SEP|||: The total input volume = the sum of the volumes of all objects. Total rise = total input volume $\div$ base area. Take out the object |||SEP|||: The water surface will drop. Descending height = volume of the removed object $\div$ bottom area. New water height = original water height – descending height.

② **First in, last out** |||SEP|||: Calculate the change after input first, and then calculate the change after withdrawal. Processed step by step in chronological order. Two-container problem |||SEP|||: Pour water from A to B. The volume poured out = A's bottom area $\times$ A's water level drops = B's bottom area $\times$ B's water level rises.

③ **Key checks** |||SEP|||: Verify at each step "Is there any overflow?" and "Is the object completely submerged?": Calculate the change after input first, and then calculate the change after withdrawal. Processed step by step in chronological order.

④ **Two container problem**: When water is poured from A to B, the volume poured out = A's bottom area $\times$ A's water level drops = B's bottom area $\times$ B's water level rises.

⑤ **critical check**: At every step, verify "Is there any overflow?" and "Is the object completely submerged?"

①

understand the situation

How many objects? In or out?

②

List known items

Base area, original water level, volume of each object

③

Calculate step by step

Calculate water level changes and new water levels according to steps

④

Verify answer

Check for overflow, units and write complete answers

□ Comprehensive examples

Example 7: The rectangular water tank is 25 cm long, 16 cm wide, and the original water height is 10 cm. A stone with a volume of 800 cm^3 is placed first, followed by an iron rod with a volume of 400 cm^3 (both are completely submerged). What is the final water height?

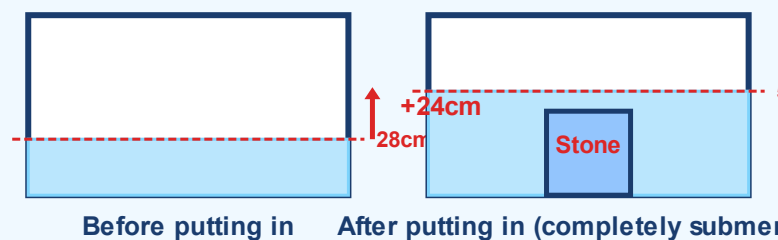
example

Example 8: The bottom area of the container is 40 cm^2 , and the height of the water is 20 cm. A block of iron with a volume of 200 cm^3 (originally completely submerged) is removed from the water. After taking it out, what is the height of the water?

Knowledge Point Four Synchronization Practice

#	Question	Difficulty	Working Space
18	The water tank is 20 cm long, 10 cm wide, and the water height is 15 cm. Place 600 cm^3 of stones and 400 cm^3 of sandbags (both completely submerged). Final water height = ?		
19	The bottom area of the container is 50 cm^2 and the water height is 12 cm. An object with a volume of 250 cm^3 is removed from the water. Water height after removal = ?		
20	The fish tank is 40 cm long, 25 cm wide, and the water height is 30 cm. (a) After placing a rockery, the water height rises to 33 cm. The volume of the rockery = ? (b) If the total height of the fish tank is 40 cm and a decoration with a volume of 3000 cm^3 is added, will it overflow?		
21	The water tank is 30 cm long, 20 cm wide, and the water height is 18 cm. First take out an object with a volume of 600 cm^3 , and then put in a new object with a volume of 900 cm^3 . Final water height = ?		

Advanced drainage method: multiple objects and removal



Water surface rise = volume of object ÷ bottom area

Review: ① Volume of object = bottom area × water level difference ② New water level = original water level + volume ÷ bottom area ③ Take out the object → the water level drops (drop = volume ÷ bottom area) ④ Multiple objects → calculate the total volume after adding up

III. Lesson Layered Synchronization Practice

Basic layer (total 5 questions, everyone must do)

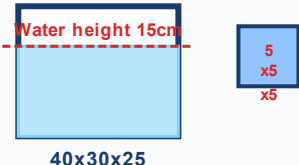
#	Question	Difficulty	Working Space
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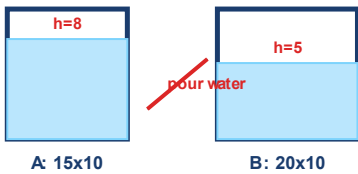
22	The base area is 25 cm^2 , the original water height is 10 cm → when an object is placed it rises to 14 cm. Object volume = ?		
23	Container capacity 800 cm^3 , original water 600 cm^3 . What is the maximum volume of objects that can be put in without overflowing?		
24	The bottom area of the fish tank is 200 cm^2 , and the water height is 15 cm. Place 1000 cm^3 of sand and gravel (completely submerged). New water high = ?		
25	The side length of the cube container is 10 cm and the height of the water is 8 cm. How many more cm^3 of water need to be added to completely fill the container?		
26	The bottom area of the measuring cup is 30 cm^2 , and the water level rises from 12 cm to 16 cm. The volume of the input object = ?		

Advanced layer (total 4 questions, choose do)

#	Question	Difficulty	Working Space
27	Rectangular water tank $30 \text{ cm} \times 20 \text{ cm} \times 25 \text{ cm}$ (height), original water volume 4000 cm^3 . Place an iron block with a volume of 2500 cm^3 (completely submerged). Will the water overflow? How much overflow?		
28	The bottom area of the measuring cup is 50 cm^2 and the height of the water is 18 cm. Three identical waves are placed and the water height rises to 21 cm. What is the volume of each waven?		
29	In a rectangular container, the water level rises by 2.5 cm after placing a 500 cm^3 iron block. What is the base area of the container?		
30	The original water in the fish tank is 40 L. After the rockery is placed, the water level rises by 5 cm. The bottom area of the fish tank is 2000 cm^2 . What is the volume of the rockery in cm^3 ? How many L is it?		

 challenge layer (total 3 questions, choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
31			

	A rectangular water tank is 40 cm long, 30 cm wide, and 25 cm high. The original water height is 15 cm. Place the iron cubes one by one with a side length of 5 cm. (a) How much will the height of the water change after adding the first iron block? (b) At which iron block is placed the water begins to overflow?		
32	 The height of water in container A (cuboid 15 cm × 10 cm) is 8 cm; the height of water in container B (cuboid 20 cm × 10 cm) is 5 cm. Pour some water from A to B so that the water levels in both containers are the same. What is the final height of the water in the two containers? (Tip: The total water volume remains unchanged, final water level = total water volume ÷ total bottom area)		
33	 A cube container with water 4 cm high. After an irregular stone with a volume of 800 cm³ is placed (completely submerged), the water height rises to 12 cm. (a) What is the base area of the container? (b) What is the side length of the container? (c) If another stone of the same volume is added, will the water overflow? (container height 15 cm)		

IV. Class after homework

Basic must-do questions (total 5 questions, column → calculate → answer sentence)

#	Question	Difficulty	Working Space
H1	The bottom area of the container is 35 cm² and the water height is 6 cm. After the object is completely submerged, the water height rises to 9 cm. Object volume = ?		
H2	The bottom area of the container is 80 cm², and an object with a volume of 400 cm³ is placed (completely submerged). How many centimeters did the water surface rise?		
H3	The water tank is 50 cm long, 20 cm wide, and the water height is 18 cm. An object is thrown in and the water height rises to 23 cm after it is completely submerged. What is the volume of the object?		

H4	Container capacity 1.5 L, original water 1 L. How many cm^3 is the remaining space?		
H5	An irregular stone is placed in a measuring cup (base area 32 cm^2) and the water level rises from 7 cm to 11 cm. Volume of stone = ?		

Advanced choose doquestion (total 3 questions,  choose do)

#	Question	Difficulty	Working Space
H6	The side length of a cube container is 15 cm and the height of the water is 6 cm. Place an iron block with a volume of 1350 cm^3 (completely submerged). Will the water overflow? If it overflows, how many cm^3 will it overflow?		
H7	The bottom area of the water tank is $45 \text{ cm} \times 20 \text{ cm} = 900 \text{ cm}^2$, and the water height is 10 cm. Place stone A (volume 800 cm^3) first and then stone B (volume 600 cm^3) until both are completely submerged. Final water height = ?		
H8	The capacity of a fish tank is 60 L, and the original water is 48 L. Place a rockery (volume 20 L, fully submerged). (a) Will the water spill? (b) If there is overflow, how much L is overflowed? (c) What will happen to the fish tank if the overflow water is poured back into the fish tank?		

V. The Lesson core common errors summary

☒ Self-examination in this hall (tick after completion)

☐ I know the pitfalls of solving each knowledge point

☐ I can complete  basic questions independently

☐ I can challenge  advanced questions

☐ I remember the formula

Review of Learning Objectives - After completing this lesson you should be able to:

☐ Identify all trap types in our hall

☐ Solve  basic questions independently (100% correct)

☐ Challenge  Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	Confusing volume and water level height SEP : Use new water height $\times$ bottom area as the volume of the object: Take the height of the new water $\times$ the bottom area as the volume of the object	Volume = bottom area $\times$ (new water level - original water level) = bottom area $\times$ water level change

2	Forgot to check for complete submersion	Drainage methods are only applicable if they are completely submerged; floating objects must be dealt with separately.
3	The units are not unified SEP : cm^3, mL and L are mixed: cm^3 is mixed with mL and L	First unify: $1 \text{ L} = 1000 \text{ cm}^3$, $1 \text{ mL} = 1 \text{ cm}^3$
4	Overflow calculation error SEP : Forgot to calculate the remaining space first: Forgot to calculate the remaining space first	Remaining space = total capacity – original water; overflow = input volume – remaining space
5	Base area calculation error	The base area of a cuboid = length $\times$ width; the base area of a cube = side length $\times$ side length
6	Missing addition for multi-object problems SEP : only the last object is calculated: Only the last object is calculated	The total volume put in = the sum of the volumes of all objects; the same is true for the total volume taken out
7	Do not write units or answer sentences SEP : only write numbers without units: Just write numbers without units	The volume must be written in cm^3 or m^3 or L; the word question must be written as "Answer:..."